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Mathematical Modelling of Wall Films Formed by Impinging Sprays

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Mathematical Modelling of Wall Films Formed by Impinging Sprays

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ABSTRACT

This paper presents a mathematical model for the prediction of the dynamic characteristics of wall films formed by impinging sprays. The model takes into account the impingement pressure due to bombardment of impinging droplets, tangential momentum transfer resulting from oblique droplet impingement on the film surface, and the gas shear force at the film surface. The general transport equations of mass, momentum and energy for wall film flows are established in the boundary-layer framework. It is shown that this set of equations can be substantially simplified if local equilibrium occurs and a dimensional analysis is performed to identify the conditions for the applicability of the local equilibrium model.

Solution of the full film equations is obtained by an efficient hybrid integral/numerical method, which allows numerical calculations to be performed in a two-dimensional framework. An implicit finite volume scheme is employed for this purpose. The methodology is first tested against some simple problems with analytical solutions. Then assessment is performed against several sources of experimental data for films formed by impinging sprays. Satisfactory agreement is obtained.

INTRODUCTION

Wall film formation resulting from spray impingement on walls is a process that occurs in both small DI Diesel engines where sprays unavoidably hit the piston walls, and in gasoline engines where port-fuel injection is employed and fuel is deposited onto the surfaces of intake valves, ports or cylinders. In all such circumstances, the film possesses the following typical features: (i) initially its mass accumulates as fuel is deposited on the walls; (ii) the surface area it covers increases continuously as a consequence of

film spreading, characterised by its moving contact line, and its continuously deforming and moving interface, under the action of a number of forces, including the shear force on the film surface due to gas swirl, body force due to wall acceleration (if present), and the tangential momentum and impact pressure due to the impinging gas jet and arriving droplets, which are especially important in the impingement region; and (iii) it exchanges heat with the walls and the surrounding gas, and consequently vapourises and mixes with the gas.

These dynamic features have a strong influence on the mixture formation and combustion processes. For instance, incomplete evaporation and burning of wall films in Diesel engines may enhance soot formation and increase unburnt hydrocarbons (see e.g. Matsui and Sugihara [1]). Also, in gasoline engines, the liquid deposited on the manifold wall or intake valve is known to affect mixture preparation quality and when it enters into the combustion chamber may cause enhanced hydrocarbon emissions, as has been observed in a transparent engine by Shin et al. [2][3]. It is therefore important to identify the major parameters which significantly influence wall film spreading and evaporation, and hence to determine the optimum operating conditions which will minimize the negative effects. This can be achieved only if the mechanisms governing spray impingement and film dynamics are understood and the phenomena can be predicted with reasonable accuracy.

The extreme complexity of the spray impingement and wall film evolution processes makes it difficult to derive information through experimental measurements. For example, the opacity of the highly dense spray in the near-wall region impedes the use of optical

techniques such as LDA (Laser Doppler Anemometer) and PIV (Particle Image Velocimetry) techniques close to the surface. Even if this difficulty is circumvented, the experimental approach is not well-suited to performing wide-ranging parametric studies, because they are time-consuming and costly. These considerations have provided the motivation for the present paper which, as a companion to an earlier publication by the present authors on droplet impingement modelling[4], is devoted to developing and assessing a mathematical model of the wall films produce by impingement.

Although literature on liquid film flows is rich, previous studies regarding the modelling of film dynamics are mainly focused on film condensation and evaporation, where no droplet/film surface impingement processes occur, see for example, References [5] to [8]. The resulting mathematical models are therefore not applicable to the prediction of wall films formed by impinging sprays. To our knowledge, the only published work that deals with the modelling of the impingement-induced wall film flows is that of Siwón and Ennaoui [9], who investigated steady and isothermal wall films formed by impinging water sprays. However, in their study, they confined their attention to two-dimensional films and also neglected the important droplet momentum sources in the film equations. For these and other reasons, their method is not applicable to the present problem.

In this study we present a wall film model which takes full account of impinging droplet contributions in the film momentum equations, addresses film heating and vaporisation, and tracks its moving contact line and deforming interface.

The remainder of this paper is organised as follows. First, we discuss the basic assumptions invoked in this study for the film modelling. We then proceed to set up the film governing equations within the boundary-layer framework. Following this, we identify the full conditions for the validity of a local equilibrium model by performing an order of magnitude analysis. We subsequently outline a solution procedure for the full and local equilibrium equations and assess it with simple test problems for which analytical solutions are available. The method is then applied to impinging-spray cases for which film measurements are available and comparisons are made with them. Finally, we summarise the main findings and conclusions.

MODEL FORMULATION

MAJOR ASSUMPTIONS - To enable the formulation of the film transport equations, the following assumptions regarding the wall film behaviour have been made:

1. The film is laminar, though it may have waves on its surface. It is thin enough for the boundary layer approximations to apply, i.e., the spatial gradients of the dependent variables tangential to the surface are negligible compared to those in the normal direction. The 'thin' film assumption can be justified by an estimation made in the present study, which reveals that the maximum film thickness is around 400µm for conditions typical of HSDI engines, even under the assumption that all the arriving droplets adhere to form a static film.
2. The incident droplets cause a local build-up of film thickness and impact pressure $p_d = (1 - q)arU_{In}^2$ ($0 < a < 1$) at the interface, where U_{In} is the normal component of the incident droplet velocity and q is the local void fraction of the gas phase. The pressure p_d is due to the loss of droplet normal momentum caused by interaction with the surface. The impacting gas also induces an impact pressure, $p_g = 1/2 r v_{gn}^2 q$.

In addition, there is, another normal force present at the interface, namely the capillary (or Laplace) pressure p_s which arises from surface tension s and the interface curvature C and is expressed as:

$$p_s = -\sigma \cdot C \quad (1)$$

where C is given by:

$$C = \frac{1}{R_1} + \frac{1}{R_2} \quad (2)$$

Here R_1 and R_2 are the principal curvature radii of the film interface. In view of the fact that the capillary force may be important at the late film spreading stages, we include it here for the sake of completeness of our formulation.

Thus, the liquid film is locally subject to a total pressure, i.e., $p_L = p_d + p_g + p_s$ at the interface. Under the boundary-layer approximations, the local pressure within the film can be taken constant across the film depth, equal to p_L .

3. The tangential momentum lost by the incident droplets provides a source of tangential momentum to the wall film.

4. Film motion is caused by spatial variations in the tangential directions in the total pressure p_L , shear at the film interface, tangential momentum sources provided by incoming droplets, and body forces.
5. The mass flux due to incident droplets may be represented by a spatially smoothly varying function, so that conventional differential operations apply.
6. The thermal radiant and Dufour energy flux in the energy equation, as well as the pressure and thermal diffusion flux in the species equations, are negligible.
7. The gaseous mixture of air and fuel-vapour behaves like an idea gas.
8. Physical equilibrium prevails at the gas/film interface, and variations of thermodynamic properties of the liquid phase, except for latent heat, are negligible.

GOVERNING EQUATIONS - we now proceed to set up the governing equations for wall films based on the above assumptions. For the sake of simplicity and clarity, we work in terms of a Cartesian coordinate system, though in the full implementation we employ a general curvilinear coordinate system. To this end, we represent two orthogonal coordinates tangential to the surface by x and y , and the coordinate normal to the surface by z . The film velocity components in these directions are respectively designated by u , v and w .

Conservation equations - Within the boundary layer approximations, the conservation equations for mass, momentum and energy take the following forms.

Continuity

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad (3)$$

Momentum

$$\begin{aligned} \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = \\ -\frac{1}{\rho_f} \frac{\partial p_L}{\partial x} + a_x + \nu_L \frac{\partial^2 u}{\partial z^2} + \frac{S_{u,d}}{\rho_f} \end{aligned} \quad (4)$$

$$\begin{aligned} \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = \\ -\frac{1}{\rho_f} \frac{\partial p_L}{\partial y} + a_y + \nu_L \frac{\partial^2 v}{\partial z^2} + \frac{S_{v,d}}{\rho_f} \end{aligned} \quad (5)$$

in which $S_{u,d}$, $S_{v,d}$ are momentum sources arising from the droplet tangential momentum loss; ρ_L represents the sum of the droplet and gas impact

pressures and the capillary pressure. The latter can be approximated as:

$$p_s \approx -s\Delta^2\delta \quad (6)$$

where δ is the local film thickness, if the terms $(\partial\delta/\partial x)^2$ and $(\partial\delta/\partial y)^2$ are negligible. These are usually true under the thin film assumptions.

Energy

$$\rho_f c_p \left(\frac{\partial T_f}{\partial t} + u \frac{\partial T_f}{\partial x} + v \frac{\partial T_f}{\partial y} + w \frac{\partial T_f}{\partial z} \right) = k_f \frac{\partial^2 T_f}{\partial z^2} \quad (7)$$

in which T_f stands for film temperature.

BOUNDARY CONDITIONS - These are required at the film surface and the wall, as follows:

Interface conditions - The conditions at gas/film interface can be derived from principles of continuity, conservation, thermodynamic equilibrium, and jump conditions (see e.g. Delhay [11]). These result in the following equations.

1. The jump mass balance equation:

$$r_f(V_f - V_s) \cdot n = r_g(V_g - V_s) \cdot n - \dot{m}_{net} \quad (8)$$

where r_f and r_g are fuel and liquid density respectively; $V_f = (u, v, w)$, $V_g = (u_g, v_g, w_g)$ and $V_s = (u_s, v_s, w_s)$ are, respectively, the film velocity vector, gas velocity vector and interface velocity vector, n is the unit outward normal vector to the interface; $\dot{m}_{net}$ is the net mass flux due to impinging droplets.

Equation (3) can be integrated with the aid of (8) to give

$$\frac{\partial \delta}{\partial t} + \frac{\partial}{\partial x} \int_0^\delta u dz + \frac{\partial}{\partial y} \int_0^\delta v dz = \frac{\dot{m}_v}{\rho_f} + \frac{\dot{m}_{net}}{\rho_f} \quad (9)$$

where $\dot{m}_v = -r_g(V_g - V_s) \cdot n$ stands for fuel vapour mass flux and takes a negative value in case of vapourisation.

2. The jump energy balance equation:

$$\dot{m}_v h_{fg} + k_f \nabla T_f \cdot n = k_g \nabla T_g \cdot n \quad (10)$$

where h_{fg} is the latent heat of vapourisation, k and T stand for heat conductivity and temperatures respectively; and the subscripts f and g refer respectively to the liquid and the gas.

3. Fuel vapour mass balance at the interface:

$$\dot{m}_v Y_{fs} - \rho_g D \left. \frac{\partial Y_f}{\partial n} \right|_s = \dot{m}_v \quad (11)$$

in which Y_f is the fuel-vapour mass fraction, and D is fuel-vapour/air binary diffusion coefficient.

4. Phase equilibrium condition:

$$T_s = T_s(p_v) \quad (12)$$

which relates the interface temperature T_s to the vapour partial pressure p_v .

5. Continuity of temperature and shear stresses:

$$T_f = T_s = T_g \quad (13)$$

$$m_f \frac{\partial u_f}{\partial z} = m_g \frac{\partial u_g}{\partial z}, \quad m_f \frac{\partial v_f}{\partial z} = m_g \frac{\partial v_g}{\partial z} \quad (14)$$

where μ_g is the gas viscosity.

Wall boundary conditions - Within the liquid-wall contact area (note that contact line is excluded), the following conditions are imposed:

1. Continuity of temperature:

$$T_f = T_w \quad (15)$$

2. No-slip condition:

$$u = v = w = 0 \quad (16)$$

3. Reduced momentum equations:

$$-\frac{1}{\rho_f} \frac{\partial p_L}{\partial x} + a_x + \nu_L \frac{\partial^2 u}{\partial z^2} + \frac{S_{u,d}}{\rho_f} = 0 \quad (17)$$

$$-\frac{1}{\rho_f} \frac{\partial p_L}{\partial y} + a_y + \nu_L \frac{\partial^2 v}{\partial z^2} + \frac{S_{v,d}}{\rho_f} = 0 \quad (18)$$

Equation (17) and (18) are used to assist in the determination of the coefficients appearing in the normalised velocity profiles adopted in this study.

CORRELATIONS FOR MASS TRANSFER - From Equation (11), it follows that evaporation rate at the film surface is given by:

$$\dot{m}_v = \frac{r_g D \left. \frac{\partial Y_f}{\partial n} \right|_s}{1 - Y_{fs}} \quad (19)$$

Using the concepts of mass transfer coefficient b_r and Sherwood number Sh_r , Equation (19) can be cast into:

$$\dot{m}_v = \frac{Sh_r r_g D (Y_{f\infty} - Y_{fs})}{l_r (1 - Y_{fs})} \quad (20)$$

Here the subscript ∞ indicates evaluation within the bulk gas adjoining the film, l_r is a longitudinal length

at position r (x, y, z) and is given as: $l_r = \frac{l_0}{\delta_0} \delta_r$, in

which δ_r is the film thickness at position r , l_0 and δ_0 are respectively characteristic longitudinal length and film thickness.

The Sherwood number Sh_r in Equation (20) is, in the absence of a better choice, evaluated from Rose's[10] correlation which was developed for two-dimensional condensations under forced-convection conditions. It has the form:

$$Sh_r = \Phi Re^{1/2} \quad (21)$$

where

$$\Phi = \frac{\Psi}{1 + 0.941 \dot{M}^{1.14} Sc^{0.93}} + \dot{M} Sc \quad (22)$$

$$\Psi = Sc^{1/2} (27.8 + 75.9 Sc^{0.306} + 657 Sc)^{-1/6} \quad (23)$$

$$\dot{M} = \frac{\dot{m}_v l_r}{\mu_f Re_r^{1/2}} \quad (24)$$

$$Re_r = \frac{V_{g\infty} l_r}{\nu_g} \quad (25)$$

$$Sc = \frac{\nu_g}{D} \quad (26)$$

Here $V_{g\infty}$ is the local tangential gas mean velocity magnitude.

LOCAL EQUILIBRIUM MODEL - It is evident that solving Equations (4), (5), (7) and (9) is by no means easy, the task can be greatly simplified if we may neglect the convective terms in the momentum equations and also assume a quasi-steady state behaviour. We now proceed to derive the conditions for these simplifications by performing an order of magnitude analysis.

We choose scaling parameters which ensure that all the dimensionless variables employed are of order unity. The two length scaling parameters are easy to select- one is the typical film thickness, δ_0 , while the other is a typical longitudinal dimension, l_0 . To determine the time scaling parameter, t_0 , we need use the mass balance equation at the interface, i.e., Equation (9), which is now rewritten as:

$$\underbrace{\frac{\partial \delta}{\partial t}}_I + \underbrace{\frac{\partial}{\partial x} \int_0^\delta u dz}_{II} + \underbrace{\frac{\partial}{\partial y} \int_0^\delta v dz}_{III} = \underbrace{\frac{\dot{m}_v}{\rho_f}}_{IV} + \underbrace{\frac{\dot{m}_{net}}{\rho_f}}_V \quad (27)$$

Since terms I, II and III are all of the order δ_0/t_0 , it follows that t_0 is of the order:

$$t_0 = \delta_0 / \left(\frac{\dot{m}_v}{r_f} + U^* \right) \quad (28)$$

where $U^* = \dot{m}_{net}/r_f$ is the mass deposition velocity. With the time scale determined, the choice of the remaining scaling parameters is quite straightforward and we arrive at the following dimensionless variables:

$$\begin{aligned} (\bar{x}, \bar{y}, \bar{z}, \bar{t}) &= \left(\frac{x}{l_0}, \frac{y}{l_0}, \frac{z}{\delta_0}, \frac{t}{t_0} \right) \\ (\bar{u}, \bar{v}, \bar{w}) &= \left(\frac{u}{l_0/t_0}, \frac{v}{l_0/t_0}, \frac{w}{\delta_0/t_0} \right) \\ (\bar{p}_\sigma, \bar{p}_{imp}) &= \left(\frac{p_\sigma t_0^2}{\rho_f l_0^2}, \frac{p_{imp}}{\rho_f U_{dn}^2} \right) \\ (\bar{S}_{u,d}, \bar{S}_{v,d}) &= \left(\frac{S_{u,d}}{\rho_f U_{dx}^2 l_0}, \frac{S_{v,d}}{\rho_f U_{dy}^2 l_0} \right) \\ (\theta, \bar{a}_x, \bar{a}_y) &= \left(\frac{T_f - T_s}{\Delta T_0}, \frac{a_x \delta_0^2 t_0}{l_0 \nu_f}, \frac{a_y \delta_0^2 t_0}{l_0 \nu_f} \right) \end{aligned} \quad (29)$$

where p_{imp} is the droplet impingement pressure ($p_{imp} + p_\sigma = p_L$); U_{dn} , U_{dx} and U_{dy} are typical impinging droplet velocity components in the normal and tangential directions, respectively; and $\Delta T_0 = [T_w - T_s]_0$ stands for the initial temperature difference between the wall and film surface.

Substitution of definitions (29) into the momentum equations gives:

Momentum

$$\frac{K}{Pr} \left(\frac{\partial \bar{u}}{\partial \bar{t}} + \bar{u} \frac{\partial \bar{u}}{\partial \bar{x}} + \bar{v} \frac{\partial \bar{u}}{\partial \bar{y}} + \bar{w} \frac{\partial \bar{u}}{\partial \bar{z}} + \frac{\partial \bar{p}_\sigma}{\partial \bar{x}} \right) = - \frac{\partial \bar{p}_{imp}}{\partial \bar{x}} \frac{U_{dn}^2 \delta_0^2 t_0}{l_0 \nu_f} + \bar{a}_x + \frac{\partial^2 \bar{u}}{\partial \bar{z}^2} + \frac{\bar{S}_{u,d} U_{dx}^2 \delta_0^2 t_0}{\nu_f} \quad (30)$$

$$\frac{K}{Pr} \left(\frac{\partial \bar{v}}{\partial \bar{t}} + \bar{u} \frac{\partial \bar{v}}{\partial \bar{x}} + \bar{v} \frac{\partial \bar{v}}{\partial \bar{y}} + \bar{w} \frac{\partial \bar{v}}{\partial \bar{z}} + \frac{\partial \bar{p}_\sigma}{\partial \bar{y}} \right) = - \frac{\partial \bar{p}_{imp}}{\partial \bar{y}} \frac{U_{dn}^2 \delta_0^2 t_0}{l_0 \nu_f} + \bar{a}_y + \frac{\partial^2 \bar{v}}{\partial \bar{z}^2} + \frac{\bar{S}_{v,d} U_{dy}^2 \delta_0^2 t_0}{\nu_f} \quad (31)$$

Energy

$$K \left(\frac{\partial \theta}{\partial \bar{t}} + \bar{u} \frac{\partial \theta}{\partial \bar{x}} + \bar{v} \frac{\partial \theta}{\partial \bar{y}} + \bar{w} \frac{\partial \theta}{\partial \bar{z}} \right) = \frac{\partial^2 \theta}{\partial \bar{z}^2} \quad (32)$$

where

$$K = \frac{\delta_0^2}{t_0 \alpha}, \quad Pr = \frac{\nu_f}{\alpha} \quad (33)$$

The important implications of the above scaled equations are almost self-evident: (i) if $K / Pr \ll 1$, the momentum equations reduce to local equilibrium form, in which viscous forces are simply balanced by body forces; and (ii) if $K \ll 1$, Equation (32) implies that film temperature simply varies linearly between the wall and interface values. Clearly, if $Pr \geq 1$, which is typical of diesel fuels, condition (ii) covers condition (i).

Once the above conditions are satisfied, the need to solve momentum and energy equations is eliminated, since we can obtain analytical expressions for the velocities and temperature. Thus the problem reduces to solving the integral mass balance equation (27) only. In some circumstances, this equation can also be solved analytically, as will be shown later when we come to test our numerical solution scheme.

METHOD OF SOLUTION

For the general non-equilibrium cases, recourse must be made to numerical methods for obtaining solutions to the film equations. In this study, we use a combination of integral and finite volume methods. This allows numerical solution to be performed in a two-dimensional framework and therefore avoids the cost and complexity of three-dimensional calculations, which in any case are scarcely warranted given the approximations involved. The overall approach, in outline form, is as follows.

1. The differential momentum and energy equations are first transformed into their integral counterparts by integrating Equations (4), (5) and (7) with respect to cross-film coordinate z .
2. Similarity is assumed in the direction normal to the surface. This allows us to introduce the following normalised velocity and temperature profiles:

$$\frac{u}{u_\delta} = f(h) = a_1 h + a_2 h^2 + a_3 h^3 \quad (34)$$

$$\frac{v}{v_\delta} = g(\eta) = b_1 h + b_2 h^2 + b_3 h^3 \quad (35)$$

$$q = 1 - (1 + \Lambda)h + \Lambda h^3 \quad (36)$$

where: $h = z/\delta$; $q = (T_f - T_s)/(T_w - T_s)$; u_δ and v_δ are velocity components at the interface; and the coefficients a_1 , a_2 , a_3 , b_1 , b_2 , b_3 and Λ are all functions of the variables x , y and t . The coefficient Λ is the temperature shape factor and it is to be determined in the course of solution, while the other six coefficients are determined by the boundary conditions.

Upon substituting the above profile expressions into the integral equations of mass, momentum and energy, there results equations of the general form:

$$\frac{\partial}{\partial t}(\phi\delta) + \frac{\partial}{\partial x}(g_\phi\phi\bar{u}\delta) + \frac{\partial}{\partial y}(h_\phi\phi\bar{v}\delta) = S_\phi \quad (37)$$

where $\bar{u}$ and $\bar{v}$ denote depth-average film velocity components in the x and y directions respectively; f stands for dependent variables 1, $\bar{u}$, $\bar{v}$ and Δ ; gf and hf represent coefficients associated with f arising from integration, and S_f is a 'source' term.

3. Equation (37) is discretised implicitly using the finite volume method. This yields, for a typical control volume shown in Figure 1 which extends from the wall to the film surface, the following discretised equation:

$$A_P f_P^n = \sum_C A_C f_C^n + A_P^o + (\bar{S}_f)_P \quad (38)$$

in which C stands for the neighbouring nodes of P ; and the superscript n and o indicate, respectively, the values estimated at the new time and old time levels.

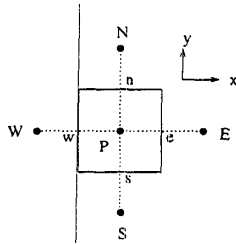


Figure 1: Layout of a typical control volume.

4. Equations (38) are solved by an iterative predictor-corrector scheme.

TEST OF THE SOLUTION SCHEME - In order to assess the accuracy of the combined integral/finite volume method just described, it is first applied to test cases for which either partial or complete analytical solutions can be sought. These do not involve spray impingement.

(a) *2-D transient film condensation* - In this case condensation of a vapour occurs on a vertical surface under conditions of negligible bulk motion and gas shear, as illustrated in Figure 2. If local equilibrium is assumed, the set of governing equations reduce to the single member:

$$\frac{\partial \delta}{\partial t} + \frac{r_f g \delta^2}{m_f} \frac{\partial \delta}{\partial x} = \frac{k_f (T_s - T_w)}{h_{fg} r_f \delta} \quad (39)$$

which is subject to the following initial and boundary conditions:

$$t = 0, \delta = 0 \quad \text{for all } x$$

$$x = 0, \delta = 0 \quad \text{for all } t$$

An analytical solution to the above can be obtained

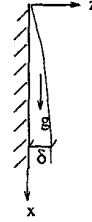


Figure 2: Schematic of transient film condensation

using a simple characteristic method proposed by Hildebrandt[12].

Comparison between the numerical solution of the full equations and the analytical one is to be made of the film thickness variation with time at a particular location using the following input data:

Parameter	Value
μ_f (Pas)	1
k_f (J/msK)	0.15
$T_s - T_w$ ($^{\circ}\text{C}$)	500
r_f (kg/m ³)	840
g (m/s ²)	9.81
h_{fg} (J/kg)	1.92×10^6

Numerical calculation of the above problem is performed using uniform grid and time increments of 0.05m and is respectively in the region $0 \leq x \leq 1$. These settings ensure negligible truncation error.

The comparison, displayed in Figure 3, clearly shows very good agreement, except in the vicinity of the time at which the analytically-determined film thickness just reaches the steady state. Here the analytical result shows a cusp, which is a consequence of using the characteristic method. In contrast, the numerical solution displays a smooth transition from unsteady to steady state, which is more plausible.

(b) *3-D transient film condensation* - Here we consider a transient condensation on a circular tube of radius R inclined at an angle α to the vertical, under conditions such that $K = 0.26$ and $K/Pr = 0.001$. We use cylindrical coordinates centred on the tube, with q measured from the upper-most generator (see Figure 4). We assume that gravity is the only external force applied on

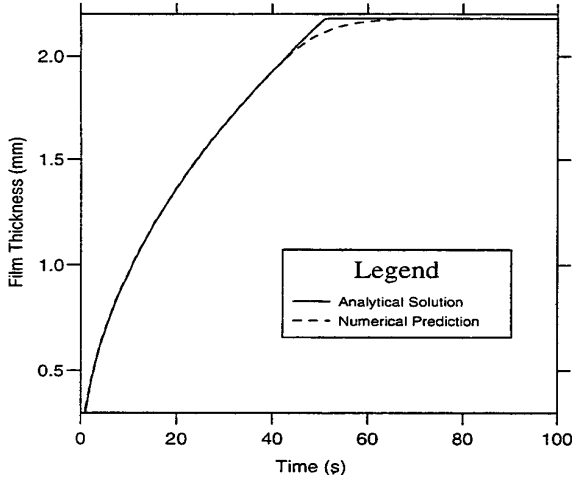


Figure 3: Comparison of numerical and analytical solutions of film thickness at $x= 1$ m.

the film, and solve the full equations, subject to the following initial and boundary conditions:

$$t = 0, d = d_0 \text{ for all } z \text{ and } q$$

$$z = 0, d = d_0 \text{ for all } t \text{ and } q$$

Here d_0 is chosen to be of very small value, ie., 1 micron.

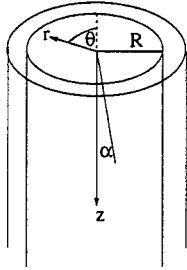


Figure 4: Coordinate system sketch

The numerical results are compared with an analytical solution which we obtained to the local equilibrium equations for the upper-most and lower-most generators $q = 0$ and $q = \pi$. The comparisons are presented in Figure 5. Two features are noteworthy: Firstly, good agreement is obtained, except in the vicinity of the cusp, for reasons explained earlier. Secondly, since the liquid inside the film has a tendency to settle towards the lower-most generator due to gravity, we see that the film thickness at $q = \pi$ is consistently larger than at $q = 0$. This also explains why the film at $q = 0$ takes longer time to reach the steady state than at $q = \pi$.

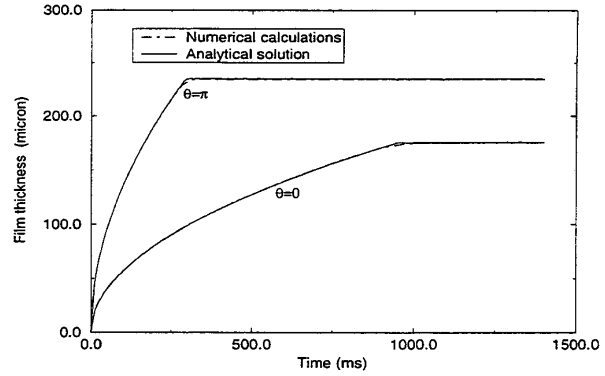


Figure 5: Numerical and analytical solutions of film thicknesses at $z= 1$ m and $q = 0, \pi$.

(c) *Verification of the criteria for local equilibrium* - As discussed earlier, we have derived general conditions for the validity of the local equilibrium model. These are now verified for a specific problem concerning again 2-D transient film condensation on a vertical plate, for which we now apply gas shear forces at the interface. We choose conditions such that $K = 0.108$ and $K / Pr = 0.042$. In the analysis, the spatial discretised region is chosen to be $0 \leq x \leq l_0$ ($l_0 = 1$ m), and the following transient inputs are used:

$$\tilde{\tau}_s(x, t) = \begin{cases} \frac{\tilde{t}}{\Delta \tilde{t}_r} \frac{\tilde{x}^{1/4}}{4} & t \leq \Delta \tilde{t}_r; \\ \frac{\tilde{x}^{1/4}}{4} & t \geq \Delta \tilde{t}_r \end{cases} \quad (40)$$

and

$$\theta_s(t) = \begin{cases} 1 + \frac{1}{2} \frac{\tilde{t}}{\Delta \tilde{t}_r} \frac{\tilde{x}^{1/4}}{4} & t \leq \Delta \tilde{t}_r; \\ 1.5 & t \geq \Delta \tilde{t}_r \end{cases} \quad (41)$$

in which all the parameters are in dimensionless form, as given below:

$$\tilde{\tau}_s(x, t) = \frac{\tau_s}{\rho_f g \delta_N}; \quad \tilde{t} = t \left[\frac{g \delta_N^2}{\nu_f l_0} \right]; \quad \tilde{x} = x / l_0$$

$$\theta_s = \frac{T_s - T_w}{(T_s - T_w)_{t=0}}; \quad \delta_N = \left[\frac{4 k_f \mu_f (T_s - T_w) l_0}{\rho_f^2 g h_{fg}} \right]^{1/4} \quad (42)$$

$\Delta \tilde{t}_r$ is chosen to be 0.1 in our analysis.

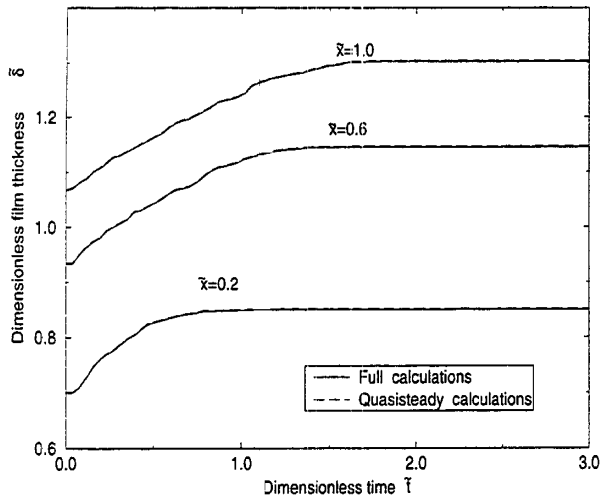


Figure 6: Variation of the dimensionless film thickness with time

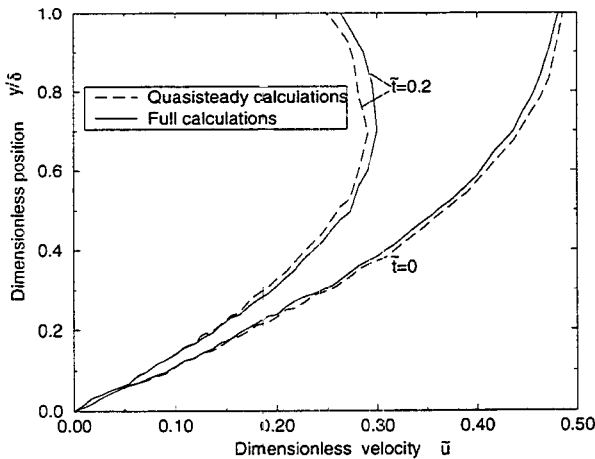


Figure 7: Dimensionless velocity distribution across film thickness

In this case both the full boundary-layer type equations and the local equilibrium ones are solved numerically. Typical results are shown in Figures 6 and 7.

Figure 6 compares variations with time of the dimensionless film thickness, $\tilde{d} = d/d_N$, at three locations, $\tilde{x} = 0.2, 0.6$, and 1.0 . It is evident that, in each location, there is an excellent match of the film thickness variations obtained from both methods.

However, the calculated dimensionless velocity profiles ($\tilde{u} = uv_f/g\delta_N^2$) at $\tilde{t} = 0.0$ and $\tilde{t} = 0.2$ for both

cases exhibit small discrepancies, as evidenced in Figure 7. These are due to the fact that the parameters K and K/Pr used herein are not sufficiently small to absolutely satisfy the local equilibrium conditions.

It is thus clear that the calculations performed for this case do provide support for our conclusions on local equilibrium conditions.

RESULTS OF IMPINGING SPRAY CALCULATIONS

To fully assess the present method under spray impingement conditions, the full model described here has been incorporated into a version of the EPISO-SPRAY code which also contains the impingement model we developed earlier.[4] Since the main features of this code and details of the impingement model are given in [4], we will not repeat them here. The code is applied to several cases previously examined experimentally and characterised by the conditions shown in Table 1. In this, the symbol S_j refers to known transient injection schedules, while cases involving supposedly constant injection pressure are indicated by the pressure level.

The main results obtained are now presented and discussed.

HARWELL EXPERIMENTS - Ereat et al [15] investigated the impingement of a spray injected normal to a flat surface in a pressure chamber containing a quiescent gas at different gas temperatures and pressures (see Cases N1 and N2 in Table 1). They used a laser sheet technique to measure the temporal variation of wall film radius, R_f (see Figure 8 for definition), which will be compared with our calculations.

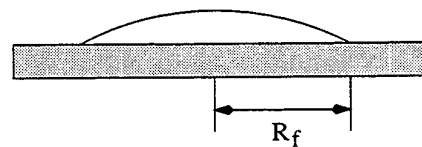


Figure 8: Sketch showing film radius definition.

Table 1: Specifications for Pressure Chamber Test Cases

Case No.	N1	N2	N3	N4	O1
Wall Distance(mm)	50	50	25	25	20
Gas Pressure(bar)	1	20	21	21	1
Gas Temperature(K)	293	473	293	293	293
Nozzle Diameter(mm)	0.32	0.32	0.25	0.25	0.25
Injection Duration(ms)	2.4	2.4	2.85	1.425	3.4
Injection Angle(Deg.)	90	90	90	90	30
Fuel Injected(mm^3 / pulse)	40	40	35	35	36
Fuel	Diesel	Diesel	Diesel	Diesel	Diesel
Injection Schedule	S_1	S_1	30 MPa	120 MPa	S_2

Figure 9 shows the comparison for Case N1, corresponding to 20°C gas temperature, and 1 bar pressure. It can be seen that the calculations are in reasonably close agreement with the measurements and capture the trend that the film radius increases monotonically and nearly linearly with time.

Figure 10 shows a similar comparison for Case N2 involving elevated gas temperature (472K) and pressure (20 bar). Again, we can see not bad agreement in general, but the measurements clearly fall below the predictions for a period starting at about 0.2ms after the impact. The measurements in fact suggest that the radius goes through a minimum during this stage. This is unexpected, since fuel injection is continuous throughout this period.

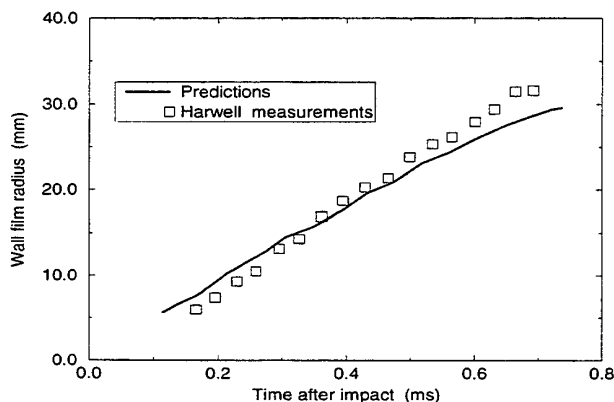


Figure 9: Comparison between predicted and measured variations with time of film radius for Case N1. (note the different scales on the two axes)

Figures 11 and 12 show respectively for Cases N1 and N2 the predicted spatial film thickness distributions

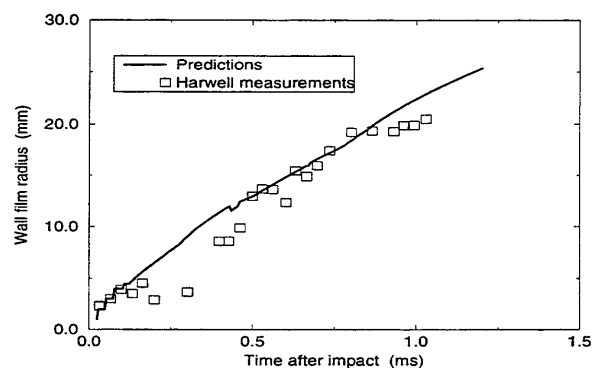


Figure 10: Comparison between predicted and measured variations with time of film radius for Case N2.

at various times (no measurements are available). Both plots show a clear mass accumulation around the film periphery in the course of film spreading. This local mass accumulation is due to the combined effects of local flow deceleration at the film edges and the relatively local larger radial velocities in the impingement zone due to the pressure increase there.

Another notable feature is that the film thickness in the impingement zone decreases with time, indicating that droplet mass flux is not enough to offset the loss caused by spreading.

It is also noteworthy that the film thickness is invariably lower in Case N2, as a consequence of much higher gas temperature, as is clear from the figures. This is due strong vaporisation of the film and droplets and less mass deposition.

EXPERIMENTS OF SAITO ET AL [13] - This investigation also involved injection in a pressure chamber under conditions closely approximating those in Diesel

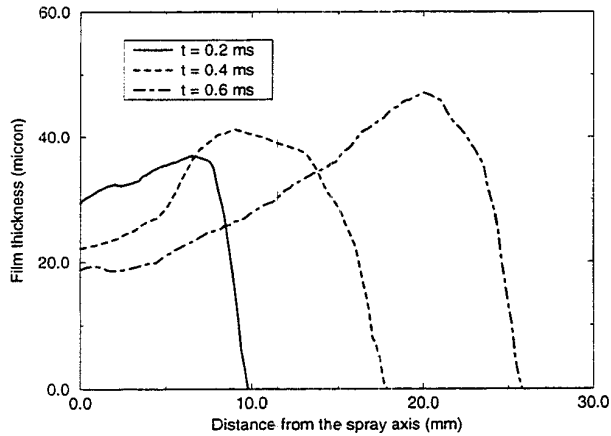


Figure 11: Predicted film thickness distributions at different times for Case N1.

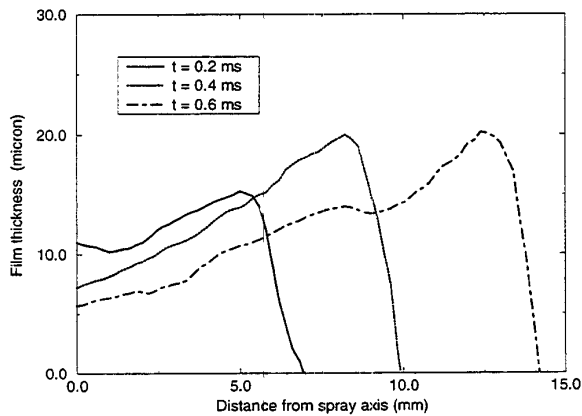


Figure 12: Predicted film thickness distributions at different times for Case N2.

engines (see Cases N3 and N4), in which both wall spray and wall film characteristics were examined. In an earlier study [4], we made comparisons with impinging spray data. Now we compare with the measurements of wall film radius and mean film thickness for two injection pressures, namely, $P = 30$ and 120 MPa.

Figures 13 and 14 show respectively the calculated and experimental variations with time of wall film radius and mean film thickness for these two cases. The mean film thickness is defined as the total film volume divided by the surface area it covers. Very good agreement is observed for film radius and satisfactory result are also obtained for the mean film thickness. The predictions clearly capture well the trend that, as the injection pressure is increased, the film spreads faster whilst the mean film thickness becomes smaller. The reasons for this are almost self-evident, since a higher injection pressure results in larger droplet and gas impingement velocities which in turn promote faster film spreading. This fact, together with the fact that higher injection pressure causes less mass to be deposited—as evidenced by the results in [4], serves to explain the above thickness behavior.

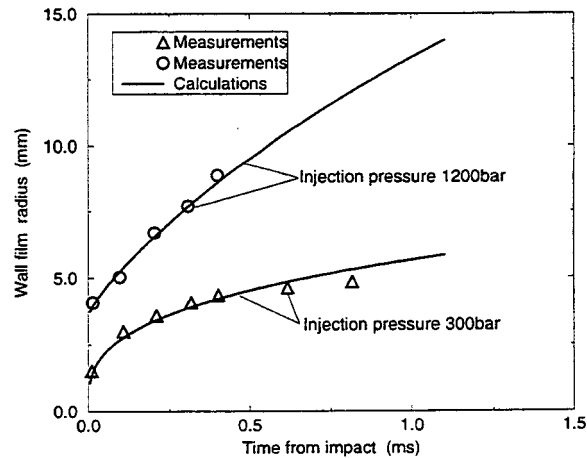


Figure 13: Calculated and measured variation with time of film radii for Cases N3 and N4.

Displayed in Figures 15 and 16 are respectively the predicted film thickness distributions at four different times: again there are no measurements to compare with. Here we can see almost the same trend as observed earlier, ie. there is mass accumulation around the film edges and the film thickness is smaller in the impingement zone. However, in Figure 16 the peak value

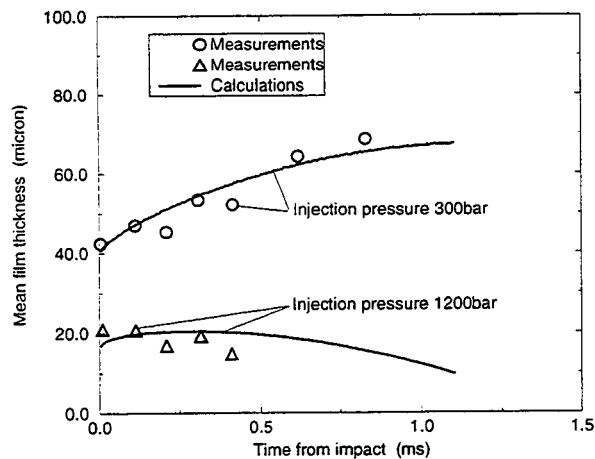


Figure 14: Calculated and measured variation with time of mean film thickness for Cases N3 and N4.

of film thickness is, unlike in the other cases, reducing with time, probably because that film spreading rate is faster.

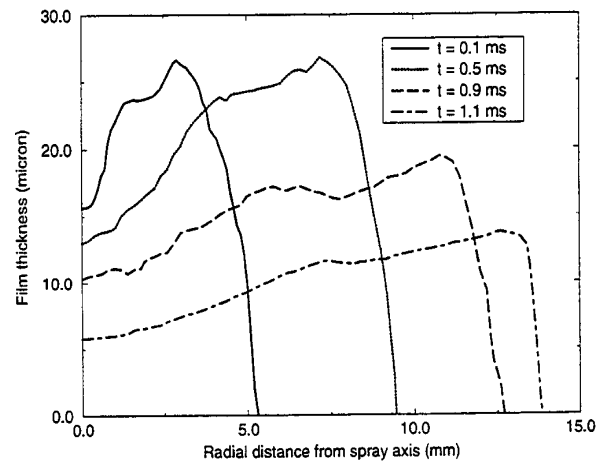


Figure 16: Predicted film thickness distributions at different times for Case N4.

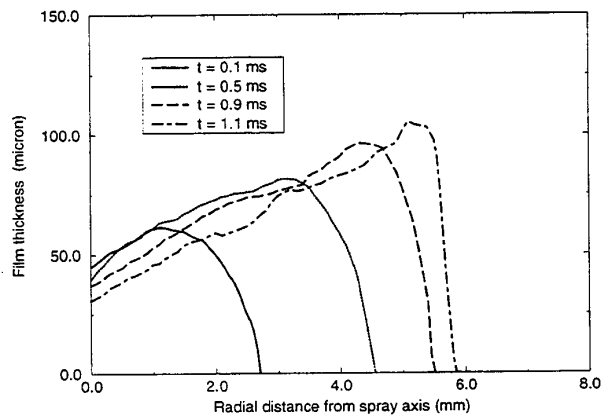


Figure 15: Predicted film thickness distributions at different times for Case N3

CNPM EXPERIMENTS - Spray impingement experiments were performed by Coghe et al [14] for the pressure chamber sketched in Figure 17 under atmospheric conditions (see Case 01). In these studies, the spray was directed obliquely at 60° onto a vertical cylindrical surface. The fluorescence/length of line technique developed in Harwell [15] was employed to measure film thickness characteristics.

Figure 19 compares the measured and predicted

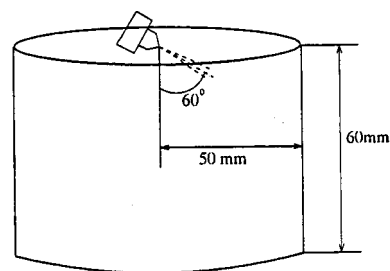


Figure 17: Sketch of CNPM experiments (Case 01).

temporal variations of the volume of liquid deposited on the wall patch ABCD, which forms part of the cylindrical impingement surface, as shown in Figure 18. The 'incidence

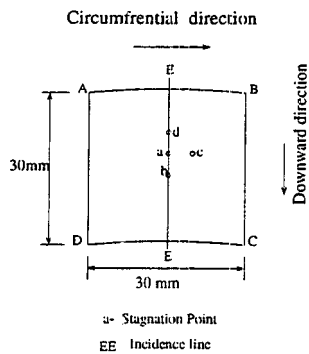


Figure 18: Wall patch of film thickness measurements.

line' in this figure denotes the line of intersection of the impingement surface with a plane formed by the spray axis and the line through the nozzle which is perpendicular to the impact surface. We observe good agreement between the predictions and the measurements from the onset of spray impingement to around 5.4 ms after start of injection. The data show a constant non-zero film volume even before impingement, which is probably indicative of the experimental uncertainty.

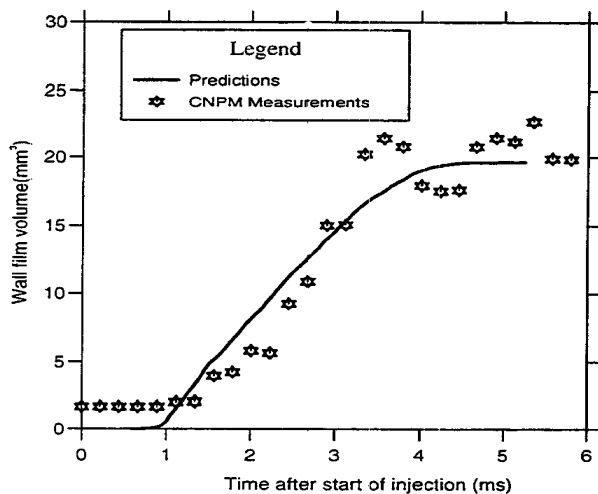


Figure 19: Predicted and measured variation with time of film volume wall patch (Case O1).

Figures 20 to 23 show for the same case predictions and measurements of the local film thickness variation at four locations a, b, c, d inside

the area of impact of the free spray cone (see Figure 18). Here point a is at the nominal stagnation point, defined as the intersection between the injection hole axis with the impact surface. Points b and d are situated, respectively, downstream and upstream of the stagnation along the incidence line. Point c lies in the line which is perpendicular to the incidence line and passes through the stagnation point. The latter three points are all 2.5 mm equidistant from point a.

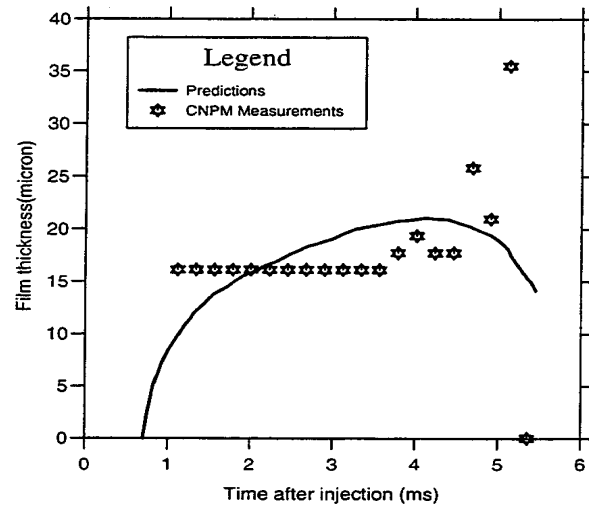


Figure 20: Predicted and measured variation with time of film thickness at point a (Case O1).

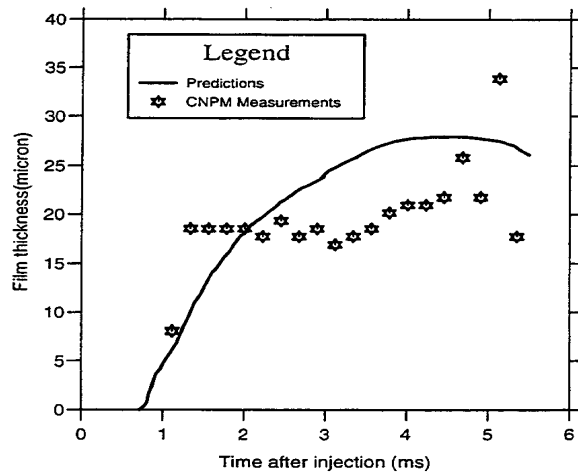


Figure 21: Predicted and measured variation with time of film thickness at point b (Case O1).

A common feature of these plots is that the predictions generally give smaller film thickness at the early

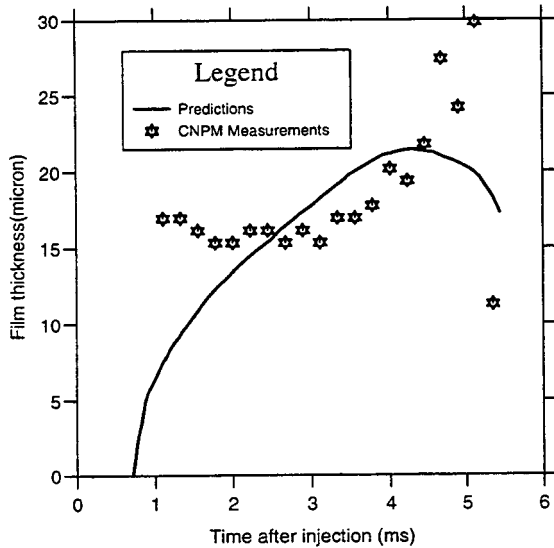


Figure 22: Predicted and measured variation with time of film thickness at point c (Case O1).

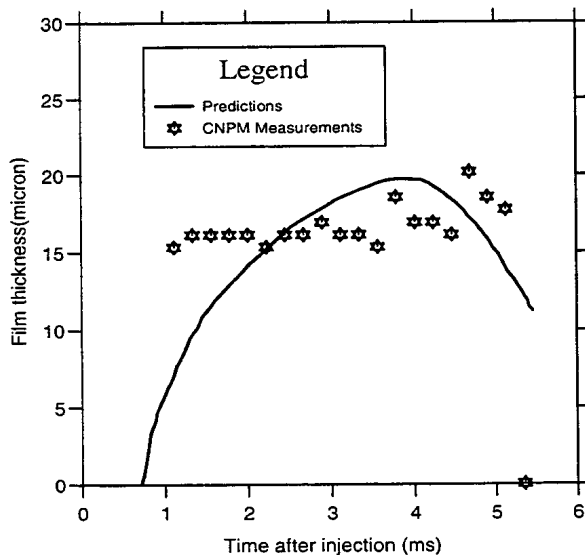


Figure 23: Predicted and measured variation with time of film thickness at point d (Case O1).

stage from the instant of impact up to 2ms after start of injection, particularly in the period immediately after impact. It can further be noted that the predictions show an expected continuous increase of film thickness as mass deposition increases; whereas the measurements produce values which are almost constant or show slight variations. The latter behaviour is particularly obvious in Figure 20 (at the stagnation point), where the film thickness is seen to be constant in the time interval from 1ms to 3.4 ms (the end of injection). This is difficult to explain, as the continuous mass deposition and impact pressure effects would almost certainly cause a dynamic variation of film thickness at this position. The explanation may lie in experimental error, especially in view of the fact that uncertainties do arise when the measured film thickness value is less than about $20\mu\text{m}$, as pointed out by Ereat et al [15].

In the period that follows, from around 2 to 5ms after start of injection (ASI), reasonable agreement can be observed between the predictions and measurements, both of which show moderate increases of film thickness with time. However, shortly after about 5ms ASI, large discrepancies occur due to the fact that the data exhibit a rapid increase (except at point d) and then decrease which the predictions underestimate. This experimental behaviour is also difficult to understand, since spray injection ceased 1.5ms earlier.

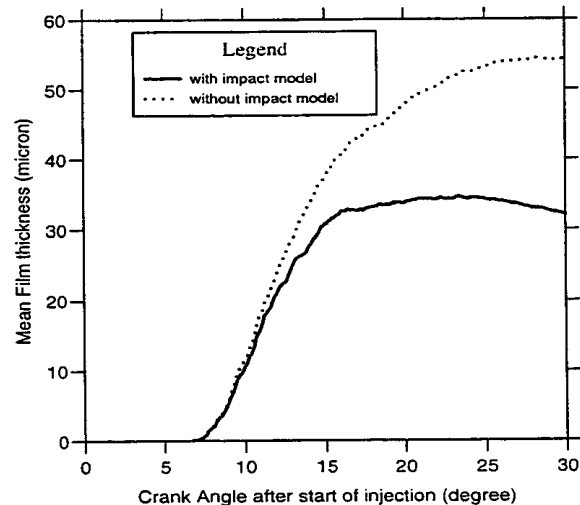


Figure 24: Predicted variation with time of mean film thickness for engine case.

ENGINE COMPUTATIONS - Finally, as a demonstration of the practical application of the method, we perform demonstration calculations on film development

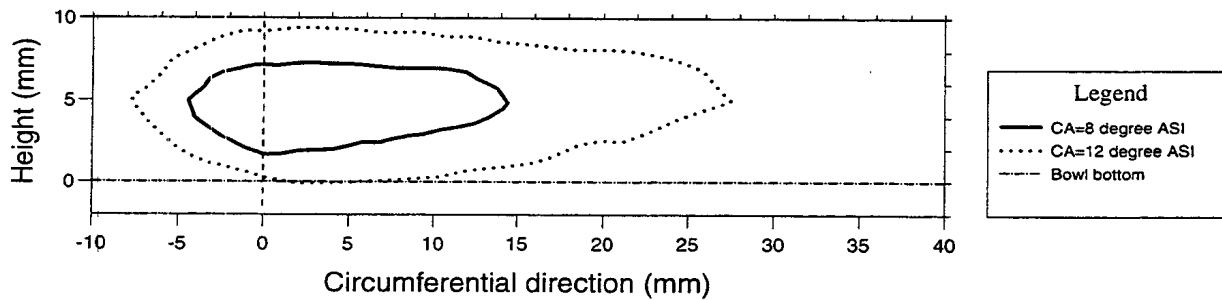


Figure 25: Predicted shapes of wall film patch on circumference of cylindrical bowl for engine case.

for an engine experiment which was used by Gutmann [16] to investigate in-cylinder spray impingement behaviour. Unfortunately, he was unable to measure the film characteristics. The engine has a cylindrical bowl and a single-hole injector which is off-centered. The pertinent dimensions and operating conditions are given in Reference [4] in which results concerning the impinging droplet characteristics for this case were presented.

Figure 24 shows the variations of mean film thickness with time using the full film model, with and without the impingement model active. In the former case the model allows for droplet splashing and bouncing, while in the latter the droplets are all assumed to adhere. This explains the differences in film thickness.

Evidence of the spreading behaviour of wall film patch in all directions as time goes on can be seen in Figure 25, which depicts the shape evolution of wall film patch at two times, plotted as the projection in a flat plane. The spreading is most significant in the tangential direction due to swirling gas drag forces.

SUMMARY AND CONCLUSIONS

A mathematical model of wall film flows has been formulated based on some simplifying assumptions. A major feature of this model is that it takes into account the effects on the film motion of the impingement pressure and tangential momentum imparted by the droplets as well as the gas shear force. This allows better representation of the film spreading process, which is important in calculating film evaporation.

The general transport equations of wall film flow were established in the boundary layer framework, and the criteria for them to reduce to local equilibrium equations were derived through an order of

magnitude analysis. The full film equations were solved by an efficient numerical method which combines integral and finite volume methods and avoids the cost and complexities of three-dimensional computations.

Validation of the model has been performed against analytical solutions and several sources of experimental data. The cases investigated covered normal and oblique impingement at ambient elevated pressures and temperatures. The quantities compared include: wall film radius, mean film thickness, film volume, and film thickness at specific locations. In all cases, comparisons made for the former three parameters showed satisfactory agreement between the predictions and the measurements. However, the predicted film thickness at specific locations show less good agreement with the measurements, although the predictions exhibit expected behaviour. Possible reasons for the discrepancies were discussed.

In conclusion, it can be said that a promising first step has been made towards the accurate modelling of engine wall film dynamics, although there is a clear need to further evaluate and extend the present method to cover more complicated cases, such as those involving turbulent flows and droplet entrainment resulting from gas-induced forces acting on the film surface - a phenomenon which may be important in port injection applications. Research in these directions is currently in progress.

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